Jayanth Krish

Electronics & Communication Engineering | **Bachelor of Technology**
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Junior Year | CGPA: 8.91/10 **Expected Graduation: Jun 2028**

Achiever's Scholarship (MIT, Manipal) *(Oct '25)*
Merit scholarship awarded to the top 5% of the B.Tech ECE cohort

WORK EXPERIENCE & RESEARCH

- **Undergraduate Researcher** | ***Mixed Signal & Computing Group (Advisor: Dr. Senorita Deb)*** *(Aug '26–Present)*
  - Architecting an autonomous, resource-constrained DDMTD core in **SkyWater 130 nm CMOS**, adapting **White Rabbit** principles for open-loop sub-nanosecond drift telemetry and clock metrology.
  - Designed an exact **modulo-1000 timebase** with direct **100.0 ps LSB quantization** without division overhead, combinational quadrant unwrapping bounded by a 500 ppm slew limit, and a programmable persistence deglitcher.
  - Synthesized standard-cell logic in **Yosys** (`sky130_fd_sc_hd`, 806 cells, 35.1% core utilization in $180 \times 115\,\mu$m) and planned physical P&R via the **LibreLane** / OpenROAD flow.
  - Performing hardware emulation on a **Xilinx Artix-7 FPGA** to evaluate code-density linearity (DNL/INL) and measurement uncertainty under **NIST-GUM**; lead author on draft for **IEEE I2MTC 2027**.
- **Electronics & FPGA Intern** | ***NCRA-TIFR, Ooty Radio Telescope*** *(May '26–Jul '26)*
  - Reverse engineered the legacy phased-array declination control link from schematics and DOS-era C code for the IBM XT/AT + **Intel 8273 SDLC multidrop bus** controlling 22 field modules (1056 dipoles).
  - Built a **TI C2000 hardware monitor** with signal conditioning to log SDLC Frames; built Python tools to decode NRZI/CRC bitstreams and isolate a live array telemetry fault.
  - Prototyped a replacement control-room backend on a Linux PC + FPGA, implementing a modular **SystemVerilog SDLC controller** (framing, bit stuffing, CRC) verified via **RTL loopback**.
- **Electronics and Power Subsystem** | ***Parikshit Student Satellite, MIT, Manipal*** *(Feb '25–May '26)*
  - Led integration for the Communication Module (UHF/VHF links, on-board data handling (OBDH), and thermal camera payload interface).
  - Developed an integrated platform for **Hardware-in-the-Loop (HIL)** testing with interfaces for payload, OBDH, and communications subsystems from system specifications to multi-layer PCB bring-up.
  - Contributed to hardware bring-up of the ADCS board, interfacing sensors, actuators, and ADCs on an **STM32** platform.
- **Hardware Engineering Intern** | ***BeePrecise (Unit of Ecosight Technologies Pvt. Ltd.)*** *(Jun '25–Jul '25)*
  - Designed and verified multi-layer **nRF52-based** IoT sensor PCBs with **ESD protection** and **DFM/DFT** compliance; built a small-batch reflow oven setup reducing assembly lead times by 1 week.

TECHNICAL PROJECTS

- **Dual-Port VRAM Framebuffer Controller**: Designed a standalone hardwired display controller in Logisim-Evolution for a 128×128 RGB-565 matrix. Implemented **interleaved 2-cycle raster timing** on a **256 KB dual-port VRAM**, bit-concatenated coordinate addressing $\{\text{Bank}, Y, X\}$, and a coordinate-comparator multiplexer for a non-destructive hardware cursor.
- **Tunable THD+N Analyzer Frontend**: Developed a variable Q-factor, tunable **Twin-Tee Notch Filter** for **audio frequencies**, achieving up to **60 dB** fundamental suppression.

POR & EXTRA-CURRICULARS

- **President**, ChipMIT: Organized technical panels with Analog Engineers from TI Dallas (150+ attendees); attended Cadence RTL-to-GDSII and TCAD workshops; won 1st place in Marvell Semiconductor Showdown. *(Present)*
- **Apprentice**, PCB Fabrication Workshop (Dr. TMA Pai Polytechnic): Executed complete prototyping flows including CNC milling, UV-cured solder masking, chemical tinning, and SMD assembly. *(Jan '25)*

TECHNICAL SKILLS

- **HDLs & EDA Tools**: SystemVerilog, Verilog, VHDL, Yosys, LibreLane (OpenLane), OpenROAD, OpenSTA, KLayout, Vivado, KiCAD, SPICE, QUCS-Studio, FlatCAM
- **Programming & Scripting**: C, Python (NumPy, SciPy, Jupyter), Bash, Make, Linux, MATLAB/GNU Octave
- **Benchtop Instrumentation**: Oscilloscopes, Logic Analyzers, Spectrum Analyzers, Signal Generators (AWGs), Bench Power Supplies, SMD Rework, HIL Testing